Secure-Us

MINOR PROJECT REPORT SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE
OF
BACHELOR IN TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING WITH SPECIALIZATION IN CYBER SECURITY
BY

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Feb, 2022

Certificate

We hereby recommend that the Project entitled ***"Secure-Us"*** worked under our guidance may please be accepted in the partial fulfillment of the requirement for the degree of "Bachelor in Technology" in the Computer Science and Engineering with specialization in Cyber Security of 'The Neotia University'. The project report in our opinion is worthy for its acceptance. During the work '**Shruti Sinha'** and **'Raj Zamal**' was found to be sincere, regular and hard working and has successfully completed the thesis work assigned to her.

……..	……..
(HOD/TIC, CSE) The Neotia University	Name of the project guide (Project Guide) The Neotia University

Declaration of Originality and Compliance of Academic Ethics

I hereby declare that this report contains literature survey and original research work done by the undersigned candidate, as part of my "Bachelor in Technology Studies".
All information in this document has been obtained and presented in accordance with academic rules and ethical conduct.
I also declare that, as required by these rules and conduct, I have cited and referenced all materials that are not original to this work.

Name: Raj Zamal
Roll No: TNU2018020100012

Project Title: Secure-Us

Signature with Date: 12/02/2022

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Project Title: Secure-Us

Signature with Date: 12/02/2022

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Chapter 1 – Introduction

Software piracy is a common occurrence, and a significant fraction of the personal computers have some pirated software installed. Cyber-criminals often use pirated software as a vector to spread malware by bundling malicious software with the pirated software. Malware detection is crucial with malware's prevalence on the Internet because it functions as an early warning system for the computer secure regarding malware and cyber attacks. It keeps hackers out of the computer and prevents the information from getting compromised.
Next, with the development of Internet technology, network security is under diverse threats. Hackers often use spam and phishing to trick users into clicking malicious URLs, the Trojans will be implanted into the victims' computers, or the victims' sensitive information will be leaked. The technology of malicious URL detection can help users identify malicious URLs and prevent users from being attacked by malicious URLs. Traditionally, research on malicious URL detection adopts blacklist-based methods to detect malicious URL. This method has some unique advantages. It has high speed, has a low false-positive rate, and is easy to realize.
Secure-Us is developed for novice users who do not have any knowledge on the above two mentioned issues of malware detection and detection of malicious urls and for those users who want a secured environment for daily usage. Secure-Us will check for malware files in the path specified by the user and ask the user regarding what is to be done to the malicious file. Secure-Us will check for malicious URLs and ask the user if it wants to continue browsing unsafely.

Background:

Malware characterized by the ability of replication, propagation, self-execution and corruption of computer systems. The corruption of computer systems can affect information confidentiality, integrity and denial of services. Replication is the important characteristic for most malware, as it ensures its existence. In some malware cases incessant replication makes exhaustion of computer resources (e.g. hard disk, RAM).
Secure-Us is a python based application which has an integrated VirusTotal API which helps the application to detect the malware in the files stored and downloaded. It prevents attacks from Trojan horse, Rootkits, Backdoors,grayware, infectious malware like virus and worm and Ransomware attack. This application provides real-time protection from the above attacks.
We might think we don't need to worry about phishing websites and the like because we only browse on safe and reputable pages such as Wikipedia.org or CNN.com. But 76% of these contain a vulnerability. To make matters even worse, 9% of sites contain a vulnerability that is serious in nature, which would allow an attacker to infect a visitor's system, download malware, or even execute code on it. So 1 out of every 11 sites we visit has such a critical vulnerability. Chances are we visit 11 new sites every week.

Secure-Us helps to provide a secure environment as mentioned above. It recognizes the malicious web pages depending on the embedded code of the webpage and warns the user for further accessing the webpage. And hence helps to secure the browsing environment of the user. Websites infected by malware can lead to scam, Malicious redirects, Browser hijacking and phishing attacks. Secure-Us warns the users if it comes in contact with a URL laced with malicious code.

Secure-Us provides an automated application for detection of malware and malicious URLs hence providing a secure environment to users who do not have any idea about these attacks and how to stop them. This application basically helps the users to automate and to ease their work, that is the continuous manual checking if the webpage URL is genuine, if it has SSL and HTTPS encryption and if has been reported before.

Aim:

The aim is to automate the process of malware detection and malicious URL detection for novice users and provide a secure environment for the system. The software will be fulfilling their requirements, so that their valuable data/information can be stored for a longer period and to prevent easy accessing and manipulation of the same by third party or hackers. Basically the project describes how to protect our files, information and personal details from the emerging risks which we face when we enter into the network and how some unwanted files can be harmful and can be used to infect others in the network.

Objective

Python based application which

- Detect Malware in the file path specified by the user
- Detect Malicious URL in the web browser
- All the detection is done on real time basis
- Provide a easy to navigate application for novice users
- Malicious URLs are predicted depending on the evaluation done on dataset using Machine Learning

In the later sections we gave the background of my project and the references we have taken from different books, research papers and websites. And gave a brief information about the different methods used and showed how they are connected using a class diagram. The reports are shown using screenshots of the application. Lastly we conclude the project and specify the future scopes of this project.

Chapter 2 – Review of earlier works

A brief look at the history of malware shows us that this malicious menace has been with us since the dawn of computing itself. According to Scientific American, the idea of a computer virus extends back to 1949, when early computer scientist John von Neumann wrote the "Theory and Organization of Complicated Automata," a paper that postulates how a computer program could reproduce itself. In the 1950s, employees at Bell Labs gave life to von Neumann's idea when they created a game called "Core Wars." In the game, programmers would unleash software "organisms" that competed for control of the computer.

From these simple and benign beginnings, a massive and diabolical industry was born. Today, according to The Anti-Phishing Workgroup, malware has infected one-third of the world's computers. The consequences are staggering. Cybersecurity Ventures reports that losses due to cybercrime, including malware, are anticipated to hit $6 trillion annually by 2021.

Anti-malware (sometimes also called antivirus) programs block and remove some or all types of malware. For example, Microsoft Security Essentials (for Windows XP, Vista, and Windows 7) and Windows Defender (for Windows 8, 10 and 11) provides real-time protection. The Windows Malicious Software Removal Tool removes malicious software from the system. Additionally, several capable antivirus software programs are available for free download from the Internet. Tests found some free programs to be competitive with commercial ones.

Typically, antivirus software can combat malware in the following ways:

- Real-time protection: They can provide real time protection against the installation of malware software on a computer. This type of malware protection works the same way as that of antivirus protection in that the anti-malware software scans all incoming network data for malware and blocks any threats it comes across.
- Removal: Anti-malware software programs can be used solely for detection and removal of malware software that has already been installed onto a computer. This type of anti-malware software scans the contents of the Windows registry, operating system files, and installed programs on a computer and will provide a list of any threats found, allowing the user to choose which files to delete or keep, or to compare this list to a list of known malware components, removing files that match.
- Sandboxing: Provide sandboxing of apps considered dangerous (such as web browsers where most vulnerabilities are likely to be installed from).

VirusTotal API

VirusTotal inspects items with over 70 antivirus scanners and URL/domain blocklisting services, in addition to a myriad of tools to extract signals from the studied content. Any user can select a file from their computer using their browser and send it to VirusTotal. VirusTotal offers a number of file submission methods, including the primary public web interface, desktop uploaders, browser

extensions and a programmatic API. The web interface has the highest scanning priority among the publicly available submission methods. Submissions may be scripted in any programming language using the HTTP-based public API.
As with files, URLs can be submitted via several different means including the VirusTotal webpage, browser extensions and the API.
Upon submitting a file or URL basic results are shared with the submitter, and also between the examining partners, who use results to improve their own systems. As a result, by submitting files, URLs, domains, etc. to VirusTotal you are contributing to raise the global IT security level.
This core analysis is also the basis for several other features, including the VirusTotal Community: a network that allows users to comment on files and URLs and share notes with each other. VirusTotal can be useful in detecting malicious content and also in identifying false positives -- normal and harmless items detected as malicious by one or more scanners.
The following Safe Browsing APIs are for non-commercial use only. If you need to use APIs to detect malicious URLs for commercial purposes – meaning “for sale or revenue-generating purposes” – please refer to the Web Risk API.

Google Extension API

The extensions APIs allow the extension's code to access features of the browser itself: activating tabs, modifying net requests, and so on. To create an extension, you assemble some resources -- a manifest, JavaScript and HTML files, images, and others -- that constitute the extension.Extensions are small software programs that customize the browsing experience. They let users tailor Chrome functionality and behavior in many ways, providing things like:

- Productivity tools
- Web page content enrichment
- Information aggregation
- Fun and games

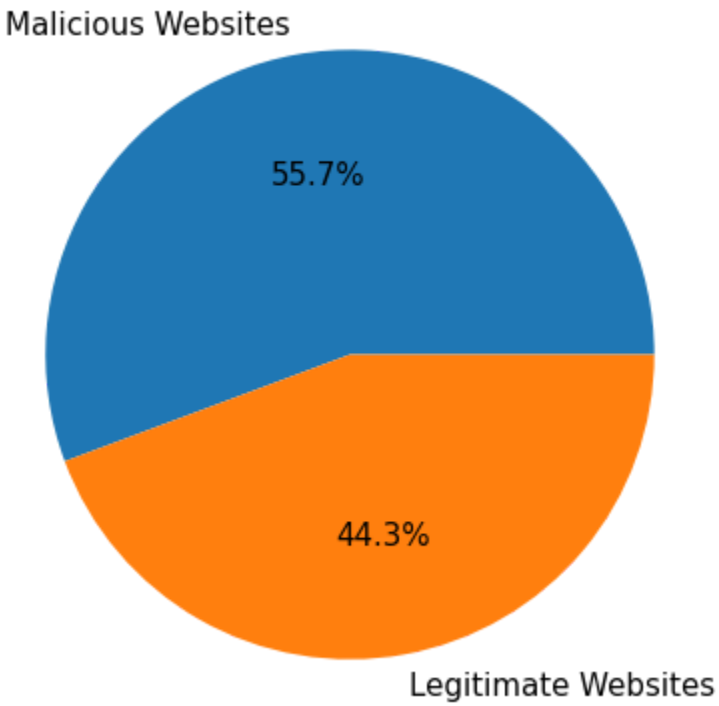
Chapter 3 – Dataset and Evaluation

MalwareChecker

We have taken the dataset from VirusShare and MalwareBazar. In this dataset we have hashes which determine if a file is malicious or not. Hashes are the unique signature of any file. In this software MD5 based hashes are used. This dataset contains around 6 lakhs hashes or malware signatures and will get updates.

Extension

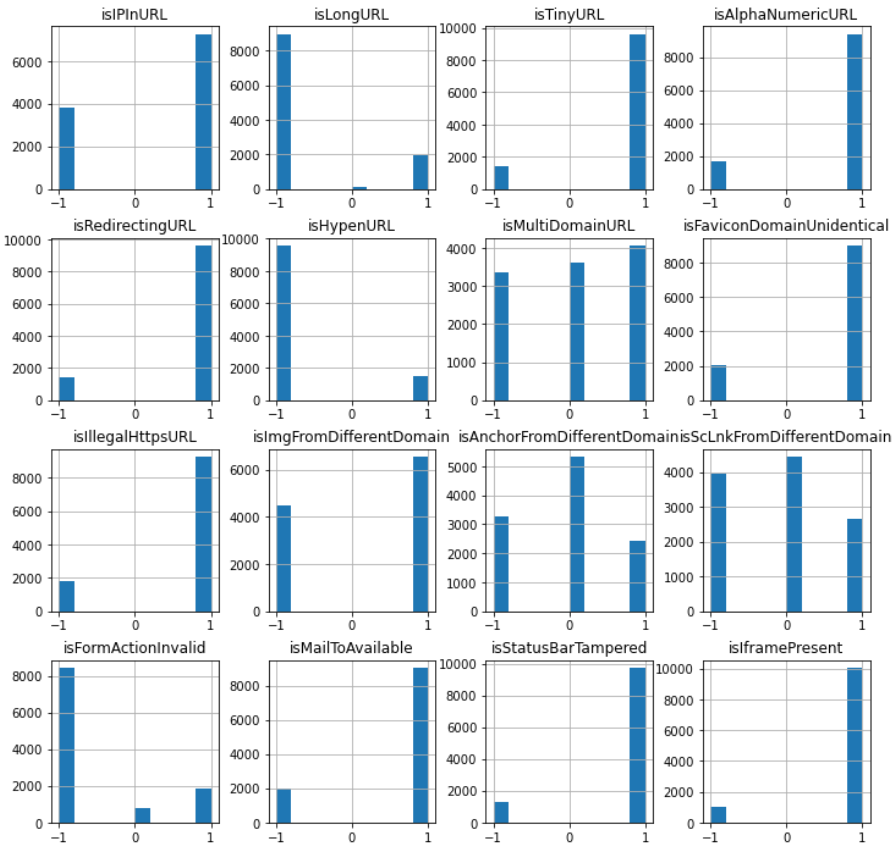
We have taken the dataset from UCI Machine Learning Repository. In this dataset, light was shed on the important features that have proved to be sound and effective in predicting phishing websites. This dataset contains 6,157 phishing and 4,898 legitimate website samples. A total of 30 different URLs, DNS, HTML, JavaScript, and External statistics based features were extracted from these websites. In this project we have used 15 such features to evaluate if the given URL is malicious or not.

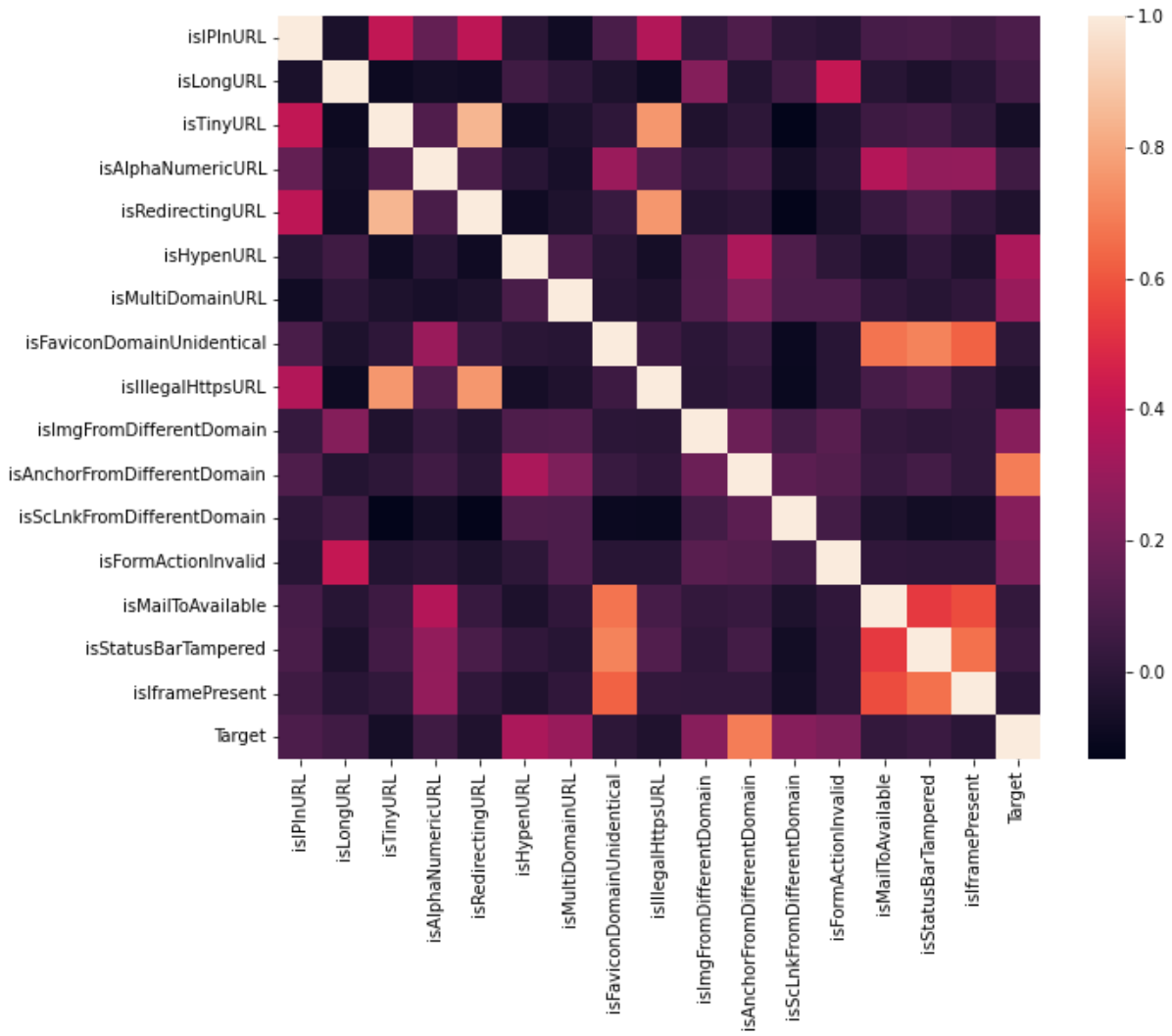


Feature and Parameters of Phishing Website Dataset

Feature	Parameters
isIPInURL	-1,1
isLongURL	1,0,-1
isTinyURL	1,-1
isAlphaNumericURL	1,-1

isRedirectingURL	1,-1
isHypenURL	1,-1
isMultiDomainURL	1,-1
isFaviconDomainUnidentical	1,-1
isIllegalHttpsURL	1,-1
isImgFromDifferentDomain	1,0,-1
isAnchorFromDifferentDomain	1,0,-1
isScLnkFromDifferentDomain	1,0,-1
isFormActionInvalid	1,0,-1
isMailToAvailable	1,-1
isStatusBarTampered	1,-1
isIframePresent	1,-1



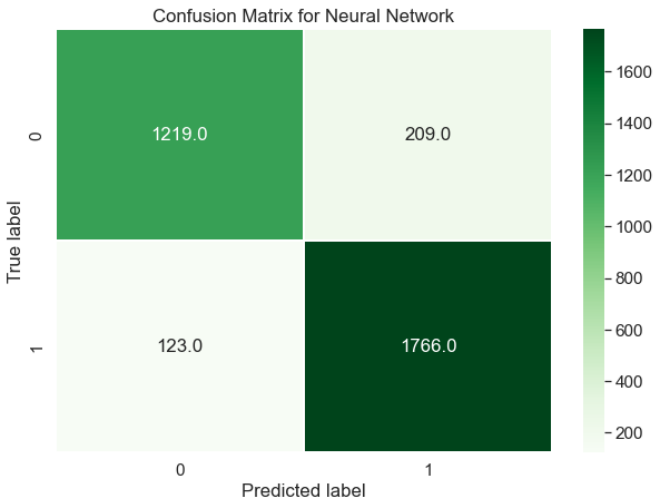


Machine Learning Models used for Evaluation

Neural Network

MLPClassifier stands for Multi-layer Perceptron classifier which in the name itself connects to a Neural Network. Unlike other classification algorithms such as Support Vectors or Naive Bayes Classifier, MLPClassifier relies on an underlying Neural Network to perform the task of classification. One similarity though, with Scikit-Learn's other classification algorithms is that implementing MLPClassifier takes no more effort than implementing Support Vectors or Naive Bayes or any other classifiers from Scikit-Learn.

	precision	recall	f1-score	support
-1	0.91	0.85	0.88	1428
1	0.89	0.93	0.91	1889
accuracy			0.90	3317
macro avg	0.90	0.89	0.90	3317
weighted avg	0.90	0.90	0.90	3317



Random Forest Classifier

Random forest, like its name implies, consists of a large number of individual decision trees that operate as an ensemble. Each individual tree in the random forest spits out a class prediction and the class with the most votes becomes our model's prediction. It is a supervised classification algorithm which can handle the classification problem even with the missing data. Its general working includes the creation of the forests which have several trees. This classifier has the capability to work for the categorical values also.

	precision	recall	f1-score	support
-1	0.90	0.88	0.89	1428
1	0.91	0.92	0.92	1889
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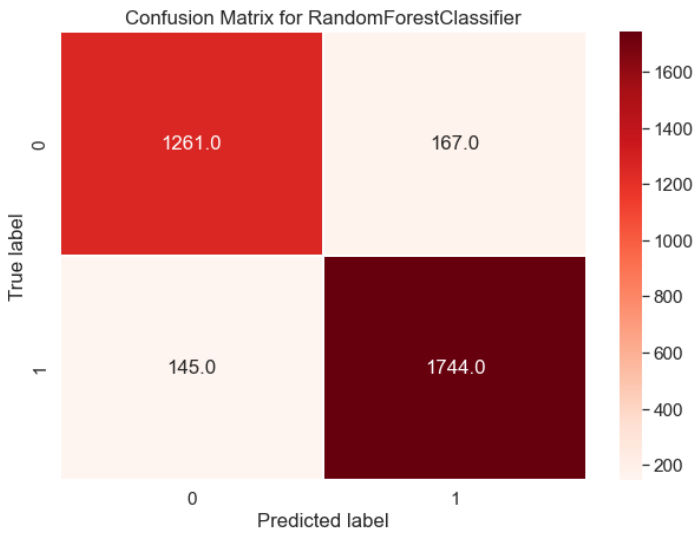
weighted avg	0.91	0.91	0.91	3317



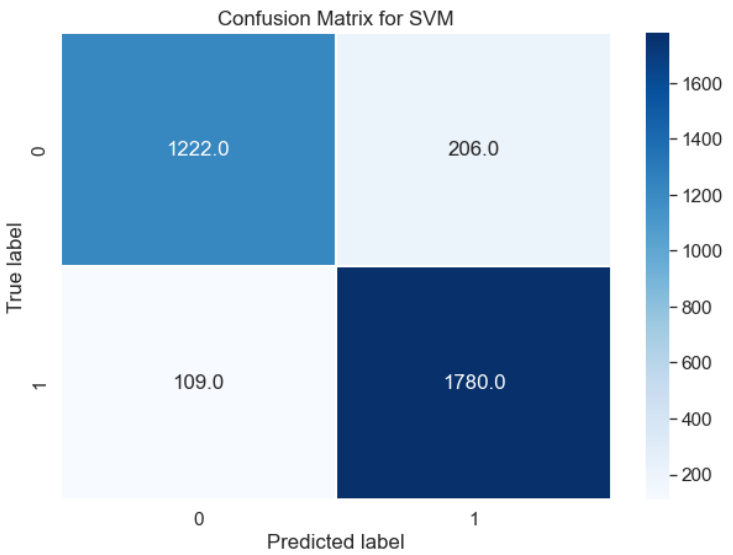
Support Vector Machine (SVM) Algorithm

Support Vector Machine (SVM) is a supervised machine learning algorithm which is applicable for both regression rather than logistic regression and classification. It makes a non probabilistic binary discriminative linear classifier. It basically creates hyper planes in infinite dimension and output or prediction is done on the basis of optimal hyperplane. So basically when the target variable is given along with labeled features then it creates a set of hyperplanes and each plane denotes each hyper plane and on the basis of that it separates into different classes.

```
              precision    recall  f1-score   support

          -1       0.92      0.86      0.89      1428
           1       0.90      0.94      0.92      1889

    accuracy                           0.91      3317
   macro avg       0.91      0.90      0.90      3317
weighted avg       0.91      0.91      0.90      3317
```



Comparison Between Models

Model used	**Accuracy(%)**
Neural Network	89.03
Random Forest	89.63
SVM	**90.05**

Chapter 4 – Methodology / Phases of Software

I started from searching how malware works and how it can be detected in files. To detect malware from the selected folder we try to get hashes of the files present and compare it with our dataset. Researched how we can extract real time URLs and check if the website is Malicious or not and what the factors that determine a website to be malicious. Searched some similar projects for reference and tried to integrate it in this project as much as possible. Python is a well-known programming language and so I chose this programming language as I will find more reference to the project.

Technology Used:

MalwareChecker

- Python 3.x
- Tkinter
- VirusTotal API

Extension

- Python 3.x
- Chrome Extension API
- Javascript, HTML and CSS

Environment

MalwareChecker

- Jupyter Notebook
- Python 3.x CLI Interpreter

Extension

- Jupyter Notebook
- Notepad++

Malware Checker

mainlike	
Initiated By	User
Status	Implemented
Type	Main
Events: 1. Checking the hash 2. If hash is changed Call hashing function	
Entry Condition	When we press the 'Start' button
Exit Condition	When we press the 'Quit' button

hashing	
Initiated By	main
Status	Implemented
Type	Controller
Events: 1. Sending the hash to virusTotal function	
Entry Condition	If hash is changed
Exit Condition	If hash is not changed
Quality Requirement	newList,newHash is valid

virustotal	
Initiated By	hashing
Status	Implemented
Type	Controller
Events: 1. Send the hash to virustotal api 2. if file is malicious ask the user to delete the file	
Entry Condition	If file is Malicious
Exit Condition	If file is not Malicious or deleted
Quality Requirement	hashed,x,vt,beforeHashed is vaild

quit	
Initiated By	main
Status	Implemented
Type	Feature
Events: 1. Quits the application	
Entry Condition	When 'Quit' button is pressed
Quality Requirement	Connected properly with tkinter

folder	
Initiated By	User
Status	Implemented
Type	Feature
Events: 1. Selects the folder to be scanned	
Entry Condition	When 'Select Folder' button is pressed
Exit Condition	When 'Quit' button is pressed
Quality Requirement	Folder path is valid

hashing2	
Initiated By	main
Status	Implemented
Type	Feature
Events: 1. Used for checking locally	
Entry Condition	When 'Yes' button is pressed
Exit Condition	When 'No' button is pressed
Quality Requirement	newList,newHash is valid

Secure-Us Extension

predict	
Initiated By	chrome extension API
Status	implemented
Type	Controller
Events: 1. Weighs evaluated from Machine Learning helps to determine if the website is malicious or not 2.Checks is a website is Malicious and returns the value to chrome extension.	

Entry Condition	Must have the data in classified format
Exit Condition	When the malicious website score f is calculated
Quality Requirement	The evaluated feature weights should be correct and the data sent to predict should be in classified format.

isIPInURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the IP address is used as an alternative of the domain name in the URL 2. If domain part has an IP Address it returns 1 else -1	
Entry Condition	It should to be able to get the real time URL
Exit Condition	It returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isLongURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Check the URL because Phishers can use long URLs to hide the doubtful part in the address bar. 2. If the URL is greater than or equal 54 characters then the URL is classified as phishing.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If the URL length <54 it legitimate if URL length>=54 and <=75 it is suspicious else phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isTinyURL

Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the URL is using URL Shortening Services	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If the URL length>20 it is legitimate else phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isAlphaNumericURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checking if the URL is using “@” symbol in the URL leads the browser to ignore everything preceding the “@” symbol and the real address often follows the “@” symbol.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	It checks for '@' if not found returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isRedirectingURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the existence and position of “//” within the URL path	
Entry Condition	It should to be able to get the real time URL
Exit Condition	It checks for the expression and if found it returns -1 which means the URL is legitimate else it returns 1 which means the URL is

	malicious
Quality Requirement	Needs to get real time URLs from the browser

isHypenURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the URL the dash symbol as dash symbol is rarely used in legitimate URLs.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If dash found in URL returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isMultiDomainURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the URL has more than one dot symbol as more than one dot symbol is rarely used in legitimate URLs.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If more than one dot found in URL returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isFaviconDomainUnidentical	
Initiated By	testdata
Status	Implemented
Type	Feature

Events: 1. Checks If the favicon is loaded from a domain other than that shown in the address bar, then the webpage.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	It the favicon is not found to be duplicate it returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isIllegalHttpsURL	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the phishers had added the “HTTPS” token to the domain part of a URL in order to trick users.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If there is https token in the URL returns 1 it is phishing URL and it returns -1 it means the URL is legitimate
Quality Requirement	Needs to get real time URLs from the browser

isImgFromDifferentDomain	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the webpage address and most of the objects and images embedded within the webpage are sharing the same domain.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If % Request URL < 22% it is a legitimate URL else if it is in the range of 22% to 61% URL is suspicious else it is a phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isAnchorFromDifferentDomain	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the <a> tags and the website have different domain names 2. Checks if the anchor does not link to any webpage	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If % URL of Anchor< 31% it is a legitimate URL else if it is in the range of 31% to 67% URL is suspicious else it is a phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isScLnkFromDifferentDomain	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if <Meta>, <Script> and <Link> tags are linked to the same domain of the webpage	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If % link in tag < 17% it is a legitimate URL else if it is in the range of 17% to 81% URL is suspicious else it is a phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isFormActionInvalid	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks the Server Form Handler(SFH) of the URL	

Entry Condition	It should to be able to get the real time URL
Exit Condition	If SHF is "about: blank" Or Is Empty it is a phishing website , if SFH "Refers To " A Different Domain it is termed as a suspicious website else it is a legitimate website
Quality Requirement	Needs to get real time URLs from the browser

isMailToAvailable	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if Using "mail()" or "mailto:" Function to Submit User Information	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If event satisfies returns 1 for phishing URL else returns -1 for legitimate URL
Quality Requirement	Needs to get real time URLs from the browser

isStatusBarTampered	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the Phishers may use JavaScript to show a fake URL in the status bar to users.	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If there is a change in status bar returns -1 it means the URL is legitimate and if it returns 1 it is phishing URL
Quality Requirement	Needs to get real time URLs from the browser

isIframePresent	
Initiated By	testdata

Status	Implemented
Type	Feature
Events: 1. Checks if the phishers make use of the “frameBorder” attribute which causes the browser to render a visual delineation	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If the URL has an iframe in its document then it returns 1 as if it is a phishing URL else -1 for legitimate URLs.
Quality Requirement	Needs to get real time URLs from the browser

getIdenticalDomainCount	
Initiated By	testdata
Status	Implemented
Type	Feature
Events: 1. Checks if the same domain is getting used and if the same domain is being used we can prompt it to stop checking	
Entry Condition	It should to be able to get the real time URL
Exit Condition	If we visit the website only once.
Quality Requirement	Needs to get real time URLs from the browser and counter should be able to keep the count a webpage has been visited

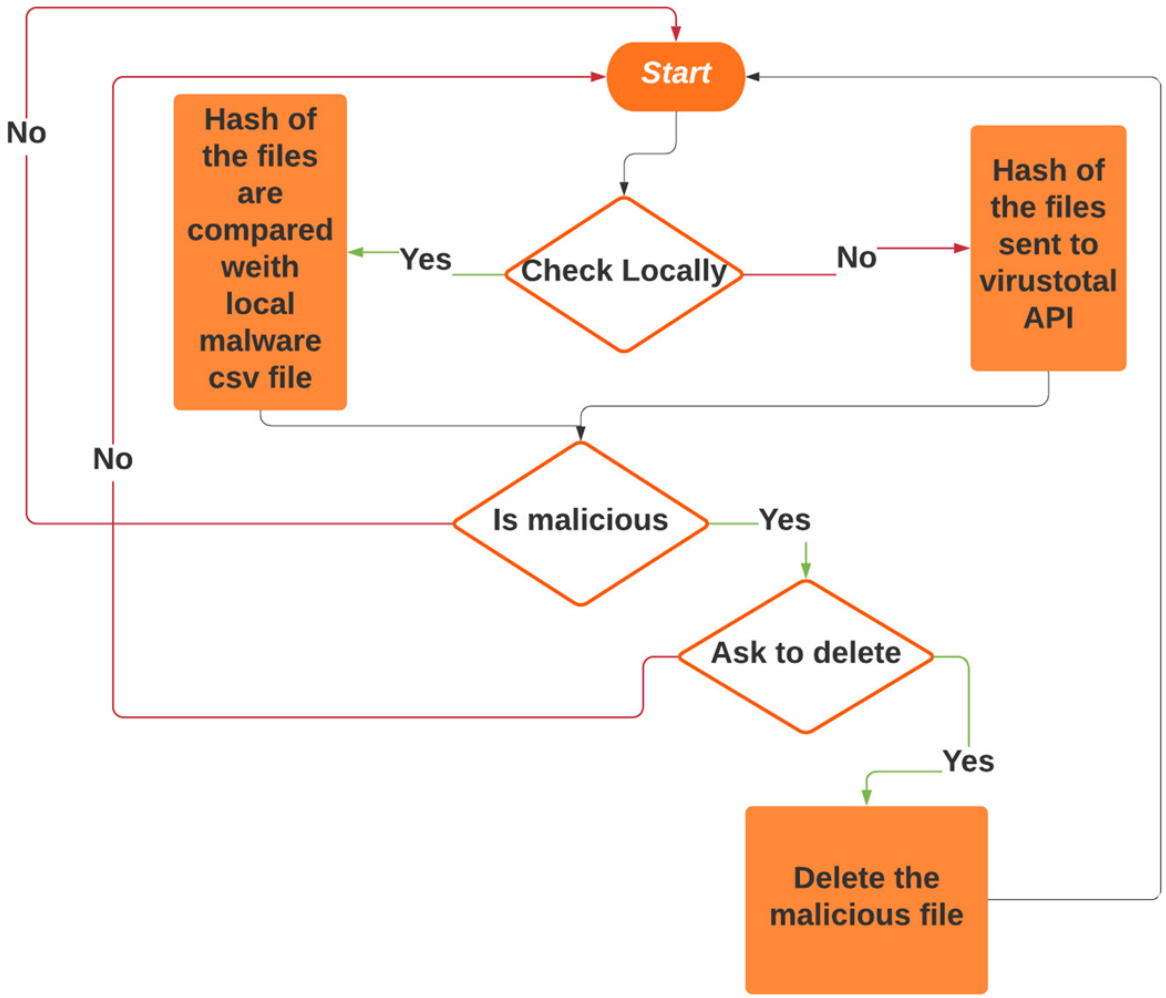
Software Development Life Cycle - Agile Model

We are using Agile SDLC to design, develop and test high quality of our software. SDLC aims to produce high-quality software that meets or exceeds customer expectations, reaches completion within times and cost estimates. We use the Agile model as it believes that every project needs to be handled differently and the existing methods need to be tailored to best suit the project requirements. In Agile, the tasks are divided into time boxes (small time frames) to deliver specific features for a release.

Agile is based on the adaptive software development methods, whereas the traditional SDLC models like the waterfall model are based on a predictive approach. Predictive teams in the traditional SDLC models usually work with detailed planning and have a complete forecast of the exact tasks and features to be delivered in the next few months or during the product life cycle.

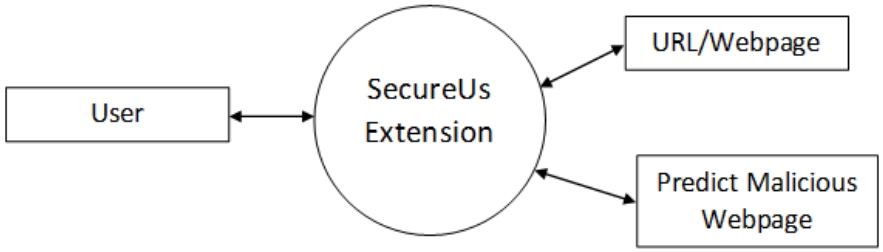
Data Flow Diagram

Malware Checker

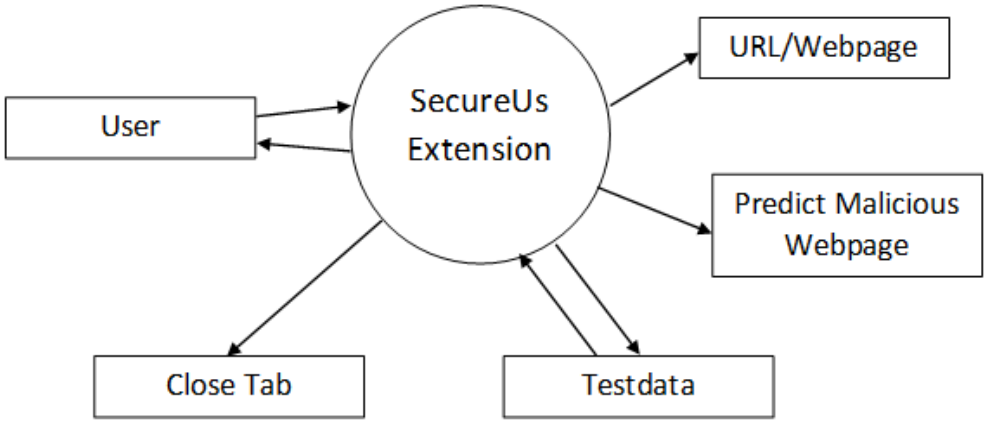


Secure-Us Extension

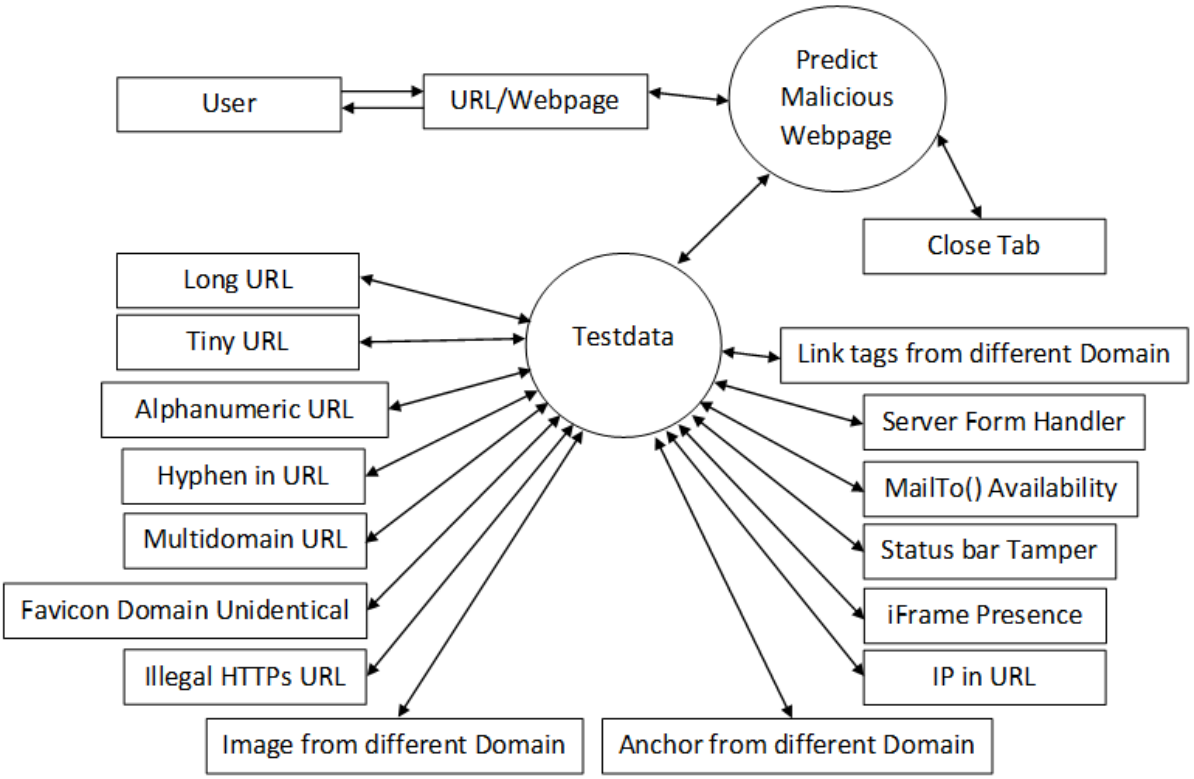
DFD Level 0



DFD Level 1



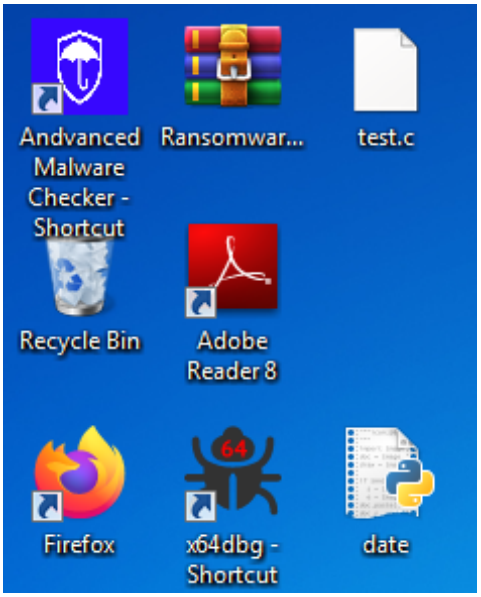
DFD Level 2

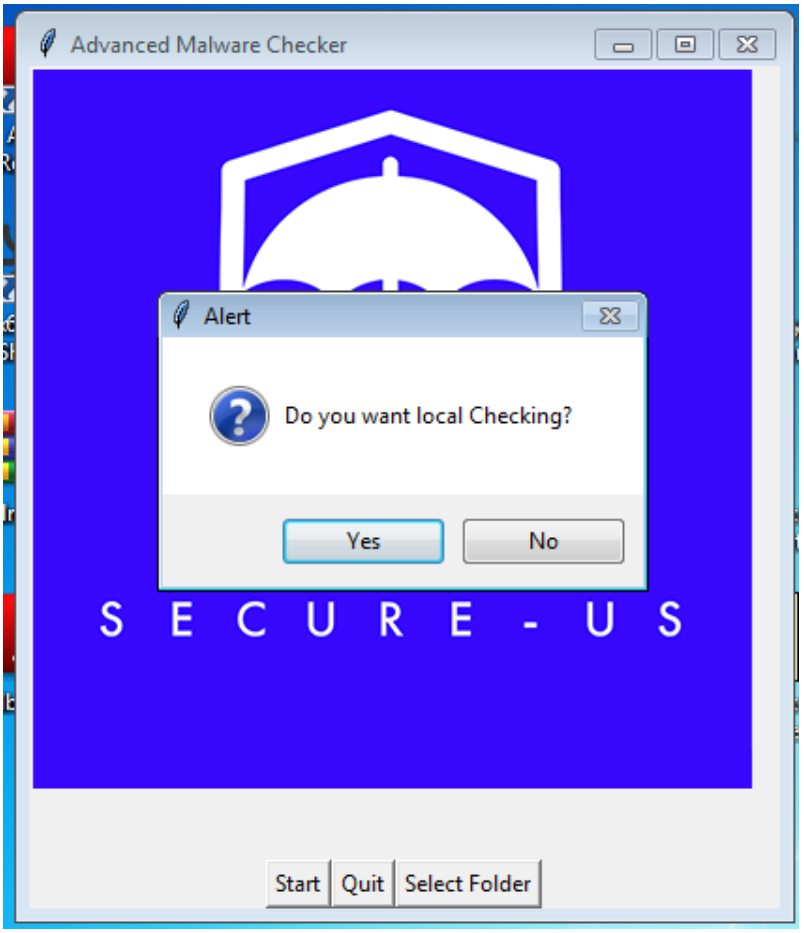


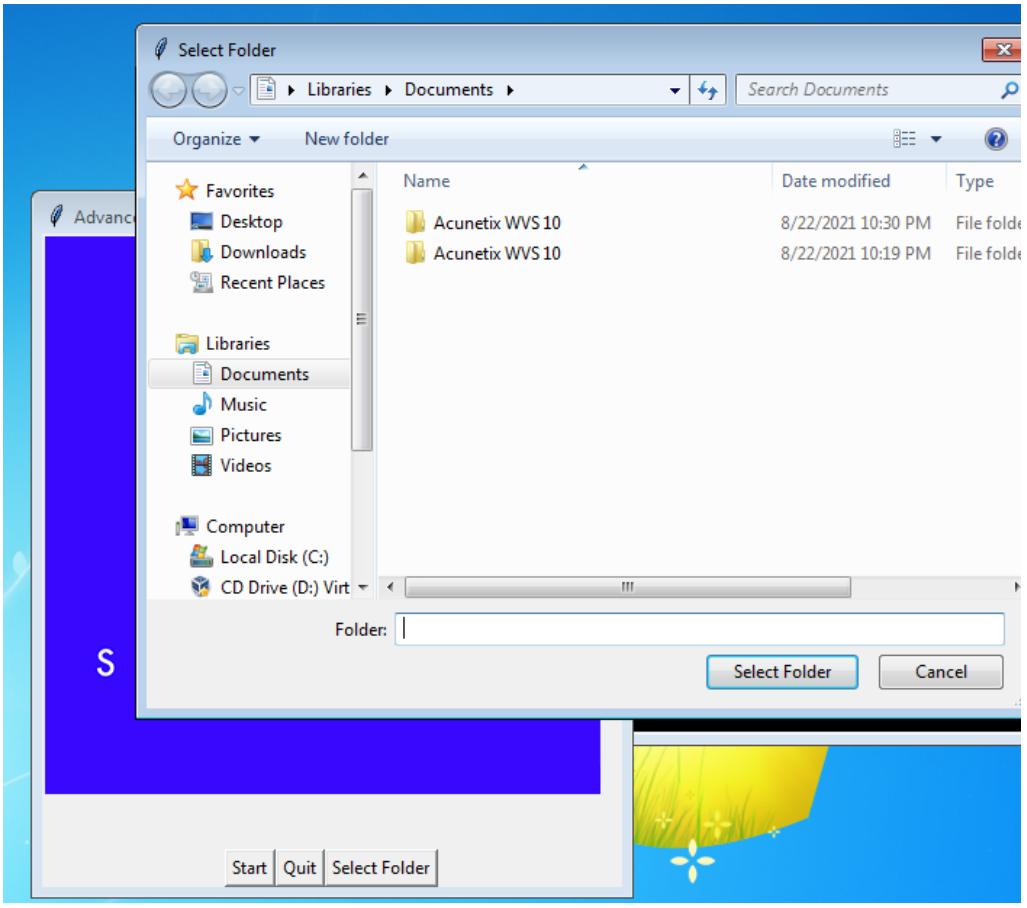
Chapter 5 – Results and Discussions

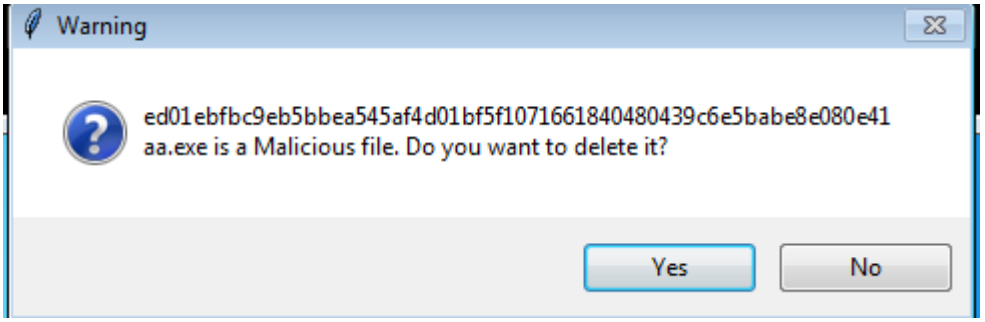
Results (Screenshots)

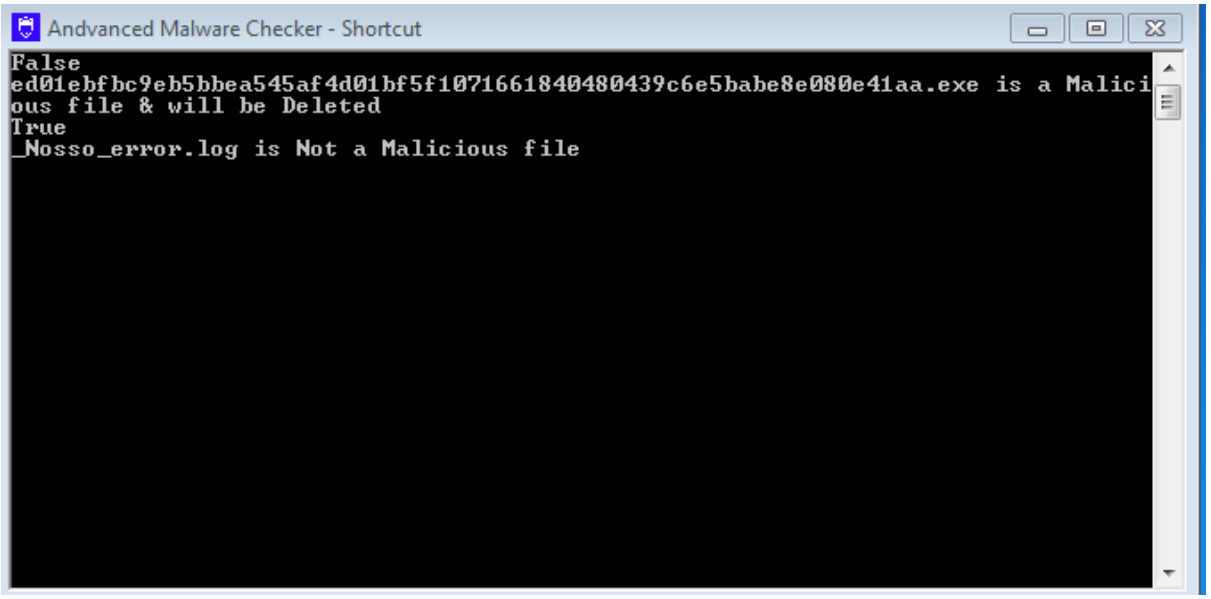
Malware Checker



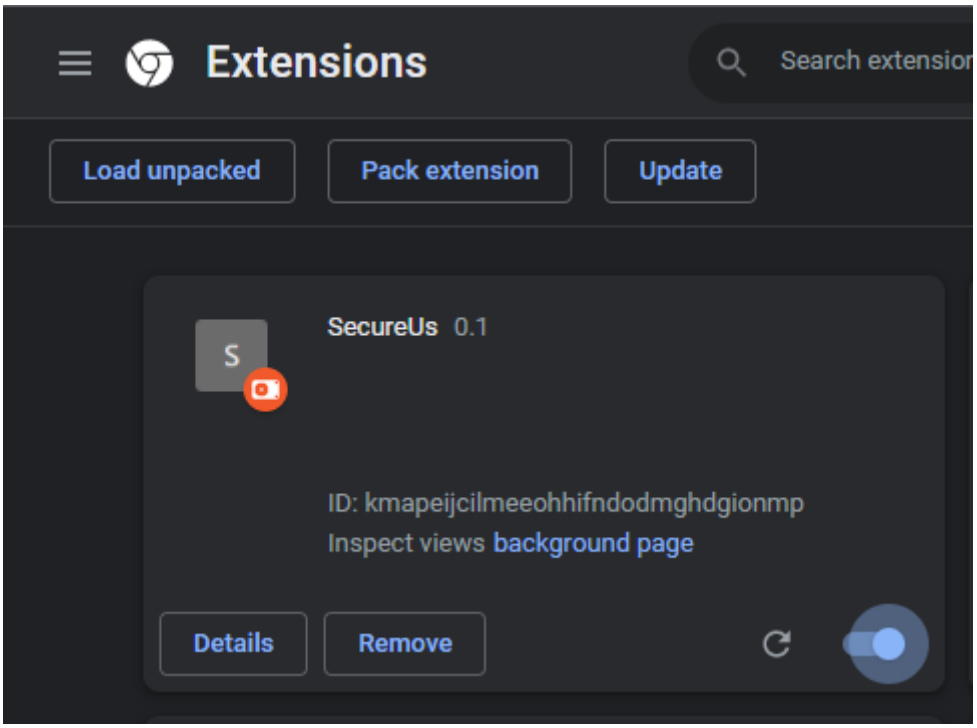

Advanced Malware Checker
Alert
Do you want local Checking?
Yes
No
S E C U R E - U S
Start
Quit
Select Folder

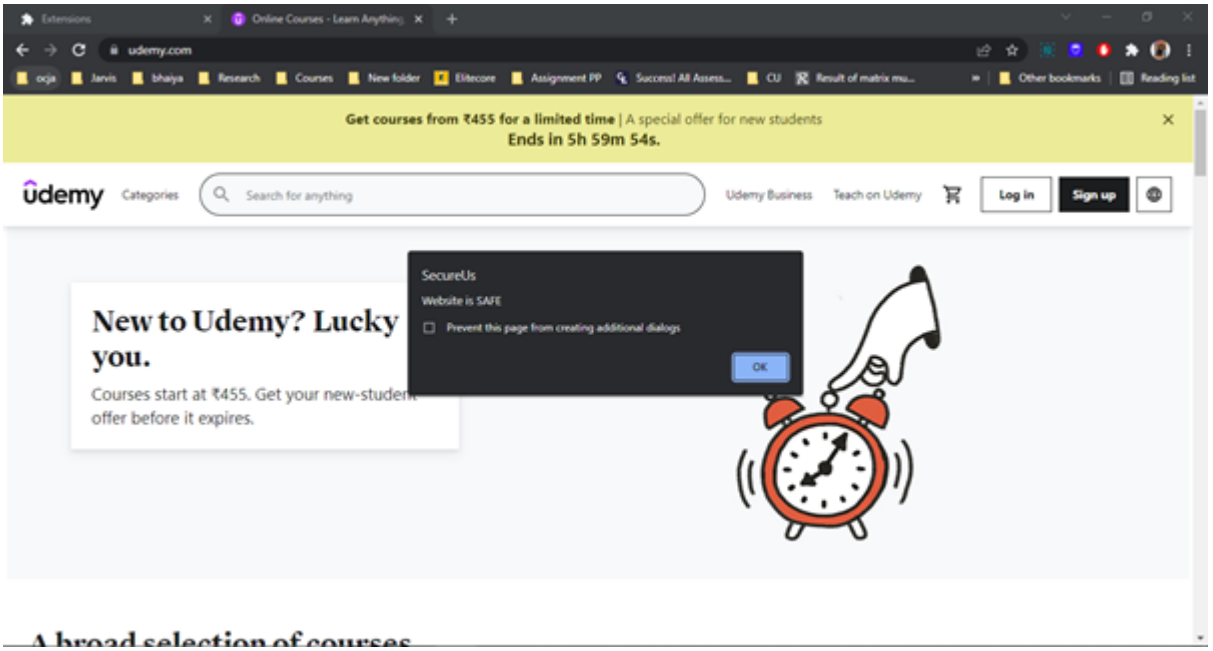

Select Folder
Libraries ▸ Documents ▸
Search Documents
Organize
New folder
Favorites
Desktop
Downloads
Recent Places
Libraries
Documents
Music
Pictures
Videos
Computer
Local Disk (C:)
CD Drive (D:) Virt
Name
Date modified
Type
Acunetix WVS 10
8/22/2021 10:30 PM
Acunetix WVS 10
8/22/2021 10:19 PM
Folder:
Select Folder
Cancel
Start
Quit
Select Folder

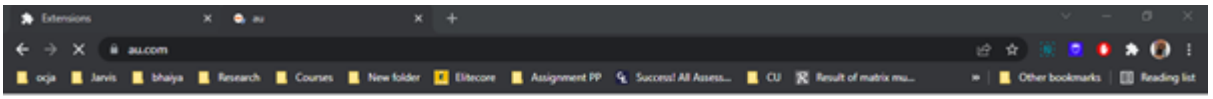


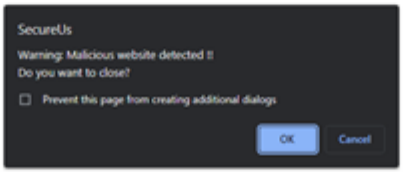


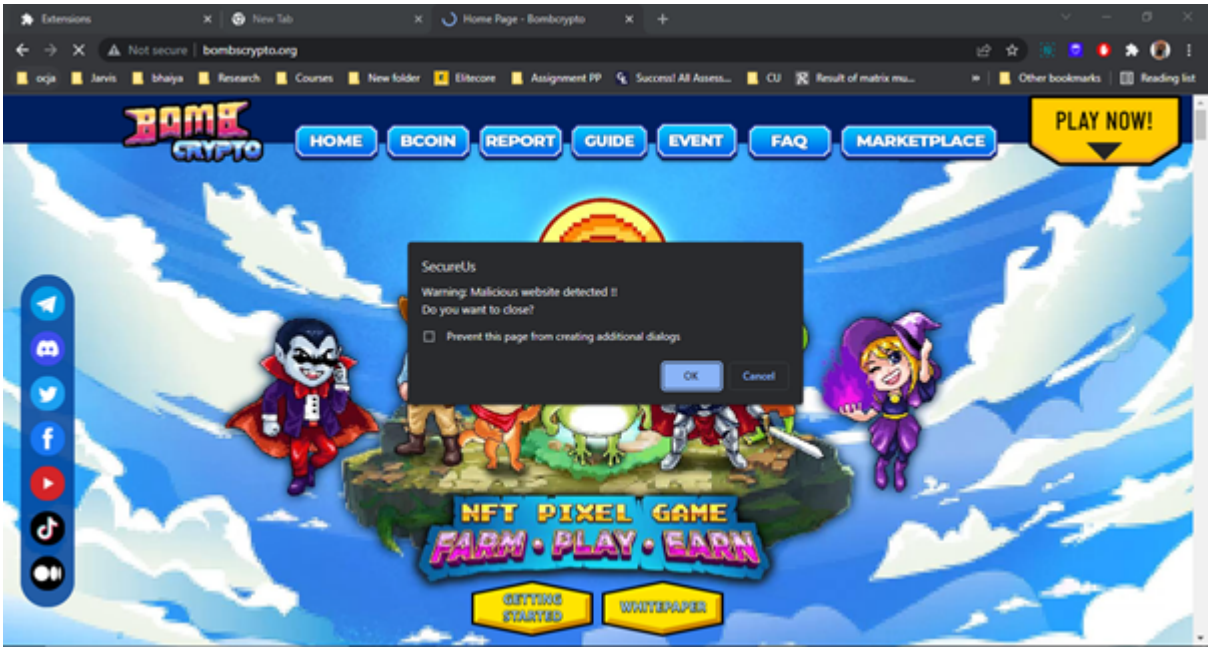
Secure-Us Extension

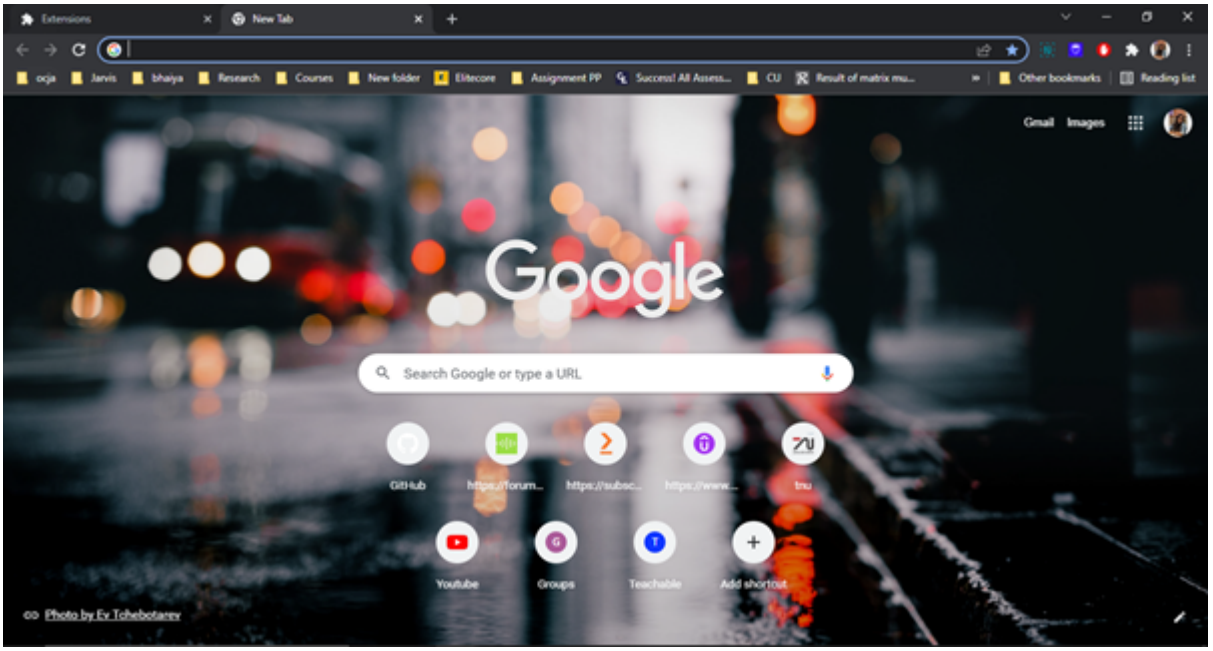


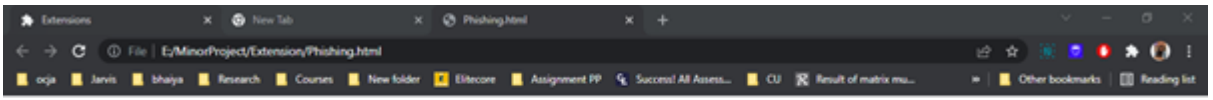

udemy.com
Get courses from ₹455 for a limited time | A special offer for new students
Ends in 5h 59m 54s.
udemy
Categories
Search for anything
Udemy Business
Teach on Udemy
Log in
Sign up
New to Udemy? Lucky you.
Courses start at ₹455. Get your new-student offer before it expires.
SecureUs
Website is SAFE
Prevent this page from creating additional dialogs
OK


au.com

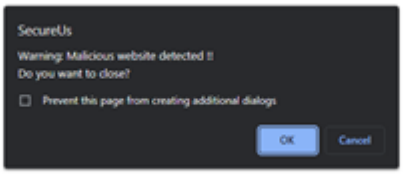

SecureUs
Warning: Malicious website detected !!
Do you want to close?
Prevent this page from creating additional dialogs
OK
Cancel

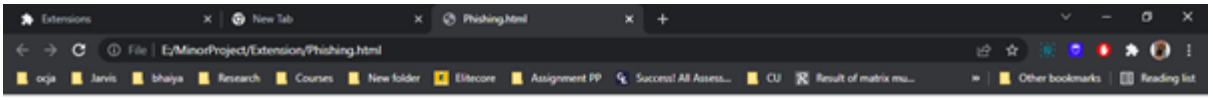

Not secure
bombscrypto.org
BOMB CRYPTO
HOME
BCOIN
REPORT
GUIDE
EVENT
FAQ
MARKETPLACE
PLAY NOW!
SecureUs
Warning: Malicious website detected !!
Do you want to close?
Prevent this page from creating additional dialogs
OK
Cancel
NFT PIXEL GAME
FARM • PLAY • EARN
GETTING STARTED
WHITEPAPER



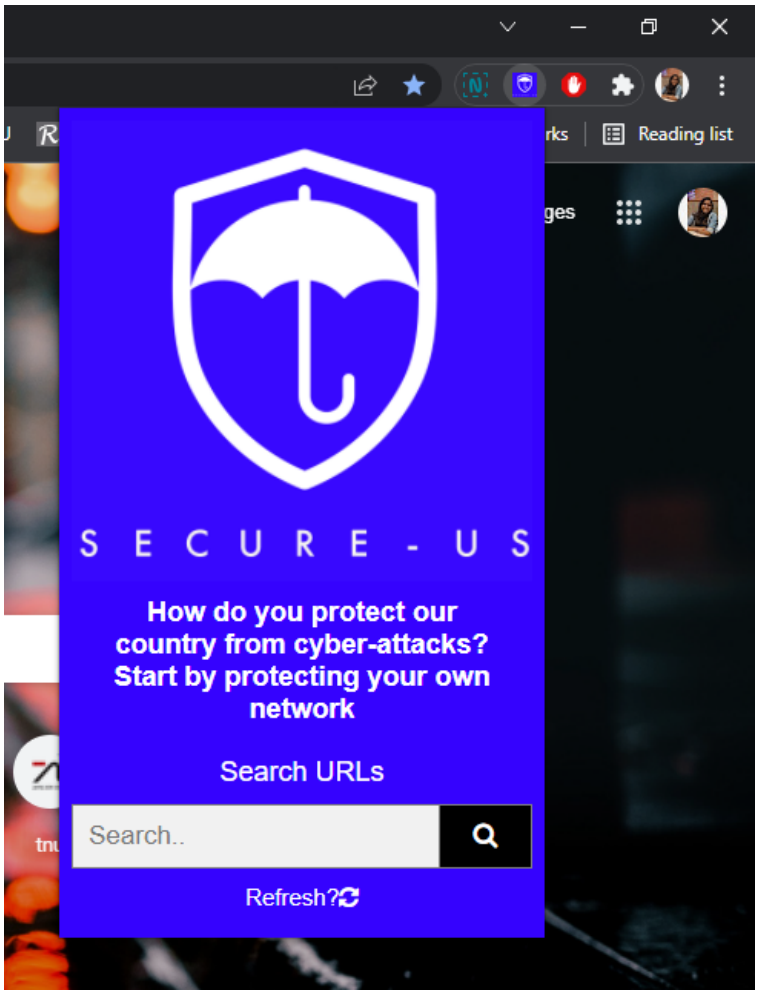


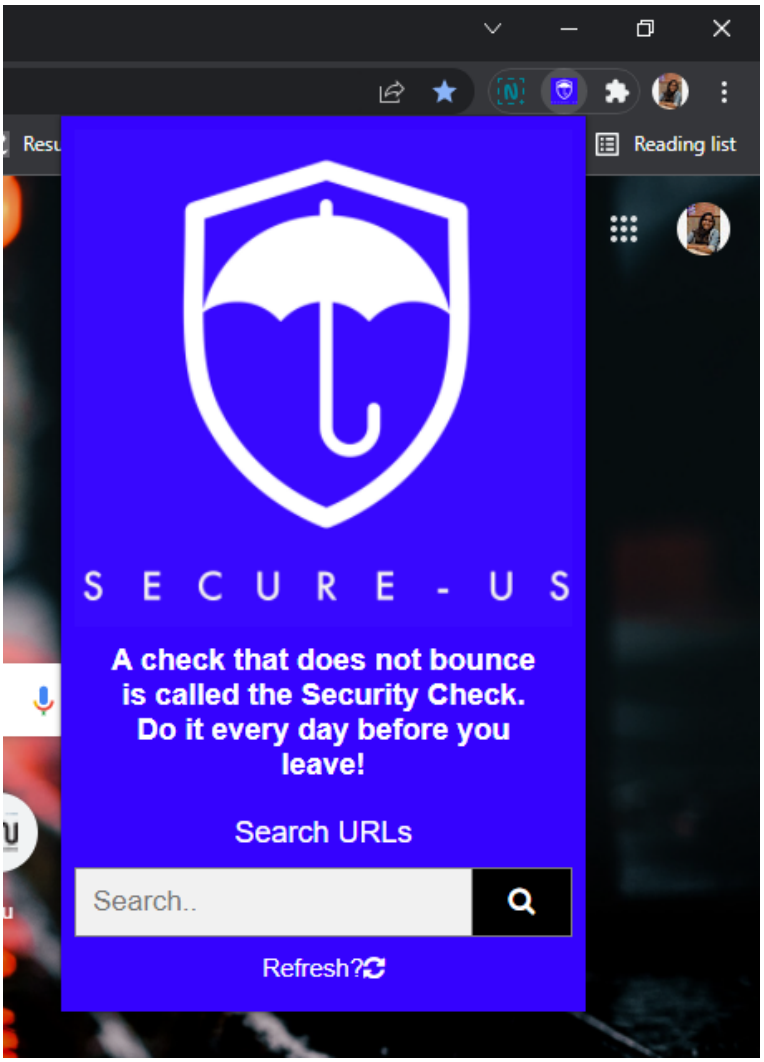
link here link here

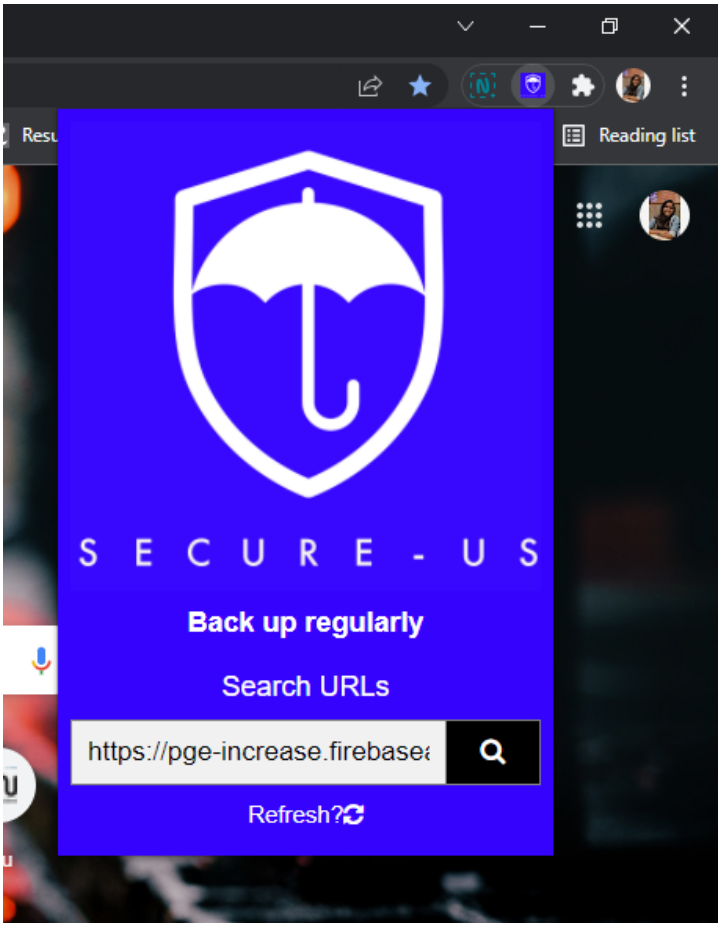


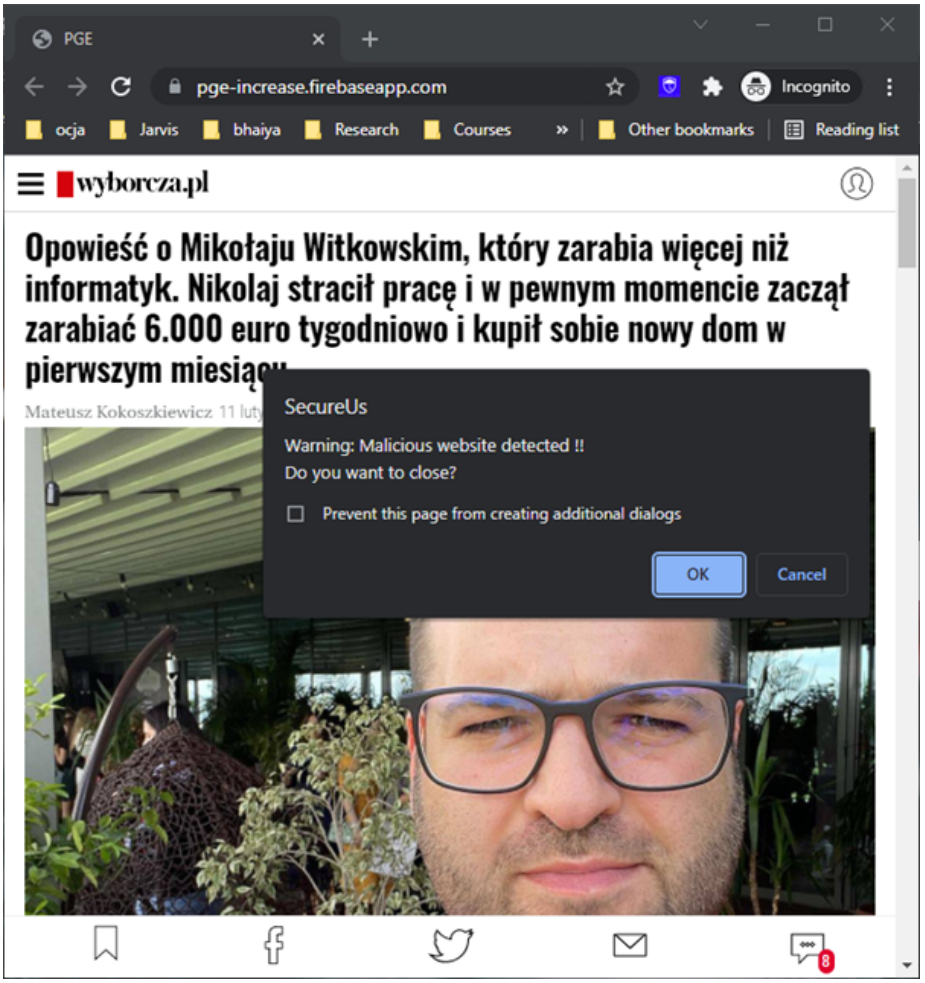


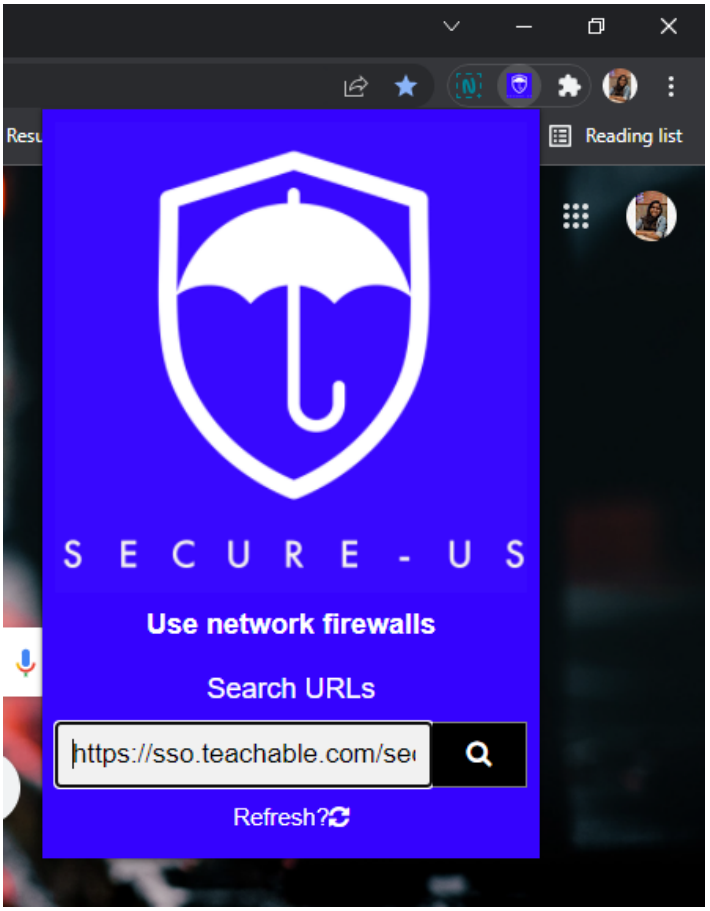
link here link here

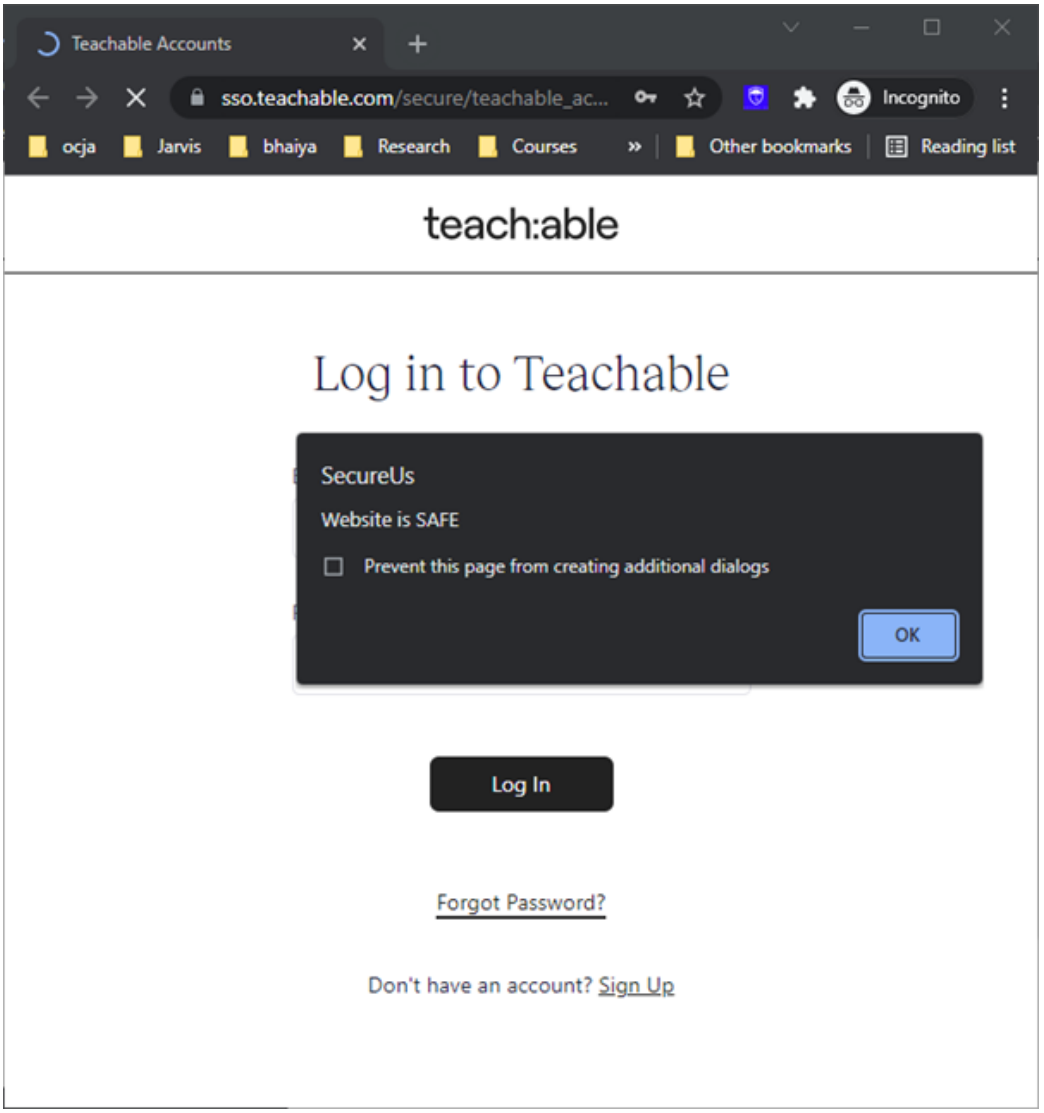

Reading list
S E C U R E - U S
How do you protect our country from cyber-attacks? Start by protecting your own network
Search URLs
Search..
Refresh?


Reading list
S E C U R E - U S
A check that does not bounce is called the Security Check. Do it every day before you leave!
Search URLs
Search..
Refresh?


Reading list
SECURE-US
Back up regularly
Search URLs
https://pge-increase.firebasea
Refresh?


PGE
pge-increase.firebaseapp.com
Incognito
ocja
Jarvis
bhaiya
Research
Courses
Other bookmarks
Reading list
wyborcza.pl
Opowieść o Mikołaju Witkowskim, który zarabia więcej niż informatyk. Nikolaj stracił pracę i w pewnym momencie zaczął zarabiać 6.000 euro tygodniowo i kupił sobie nowy dom w pierwszym miesiącu
Mateusz Kokoszkiewicz
SecureUs
Warning: Malicious website detected !!
Do you want to close?
Prevent this page from creating additional dialogs
OK
Cancel


Reading list
S E C U R E - U S
Use network firewalls
Search URLs
https://sso.teachable.com/se
Refresh?


Teachable Accounts
sso.teachable.com/secure/teachable_ac...
Incognito
ocja
Jarvis
bhaiya
Research
Courses
Other bookmarks
Reading list
teach:able
Log in to Teachable
SecureUs
Website is SAFE
Prevent this page from creating additional dialogs
OK
Log In
Forgot Password?
Don't have an account? Sign Up

Discussion

- Needs continuous internet connection to connect to the API
- Scans only files in selected folders
- Extension can only works in Chrome, Edge and Opera
- Dependent on VirusTotalAPI and Chrome Extension API

Chapter 6 – Conclusion and Future Scope

Conclusion

As technology is enhancing day by day and as it is becoming an integral part of our lives, security has become a necessity. This project is only a humble venture to satisfy the security needs of a user who is novice in technology. Our aim is to provide a user with various tools to enhance the security of their system.User friendly coding has been adopted. This package shall prove to be a powerful package in satisfying all the requirements of the user. The objective of software planning is to provide a secure system to the user and increase awareness while coming in contact with the network.

At the end it is concluded that we have made effort on following points:

- Successfully detecting malware files using the unique hashes of these files
- Successfully check the malicious and legitimate URLs we visit in our browser.
- Use the extension in 3 different browsers like Microsoft Edge, Google Chrome and Opera.
- We can assign folders and paths where we want to assign a malware check.

Future Scope

It can be summarized that the future scope of the project circles around maintaining information regarding:

- To build a software to detect if a software has malware without using hashes.
- To build an API to provide security to devices with less computational power.
- To add more features to check if the given URL is malicious or not for more accurate results.
- To publish the extension in chrome for more users.
- To provide visual analysis for better understanding.Improve GUI for easy usage

The mentioned points are the enhancements which can be done to increase the applicability and usage of this project. Here we can detect malicious files and webpages. Also, as it can be seen that now-a-days due to pandemics and more usage of systems the security of a system is a necessity. Enhancements can be done to increase detection of Security and a safer environment for novice users

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